

Power Budget

Team Number: 211

Project Name: EGR 304 Thing

Team Member Names: Levi, Michael, Hafsa, Kelton

Version: 1

All Major Components	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	Solenoid	FIT0617	+12 - 24V	1	500	500	mA
	Speaker	50CS08FH-1	+5 - 12V	1	300	300	mA
	PSoC™ 4 BLE Module	PIC18F57Q43-CNA	(range)	1	200	200	mA
	MOSFET	AOTF2618L	+5 to -5	1	100	100	mA
	9V regulator	L7805CV	+5V - 35V	1	1000	1000	mA
+12V Power Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	Solenoid	(full part number)	+12 - 24V	1	500	500	mA
	Subtotal					500	mA
	Safety Margin					25%	
	Total Current Required on +12V Rail					625	mA
Regulator	+12V regulator	AOTF2618L	+12V - 35V	1	1000	1000	mA
	Total Remaining Current Available on +12V Rail					375	mA

<i>+5V Power Rail</i>	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	PSoC™ 4 BLE Module	(full part number)	(range)	2	200	400	mA
	Speaker	(full part number)	(range)	1	100	100	mA
						0	mA
	Subtotal					500	mA
	Safety Margin					25%	
	Total Current Required on +5V Rail					625	mA
Regulator	+5V Regulator	LM7805CV	(range)	1	1000	1000	mA
Total Remaining Current Available on +5V Rail						375	mA

<i>External Power Source 1</i>	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit
Power Source 1 Selection	Plug-in Wall Supply	(full part number)	110VAC	+24V	5000	5000	mA
Power Rails Connected to External Power Source 1	+12V regulator	LM7812	+12V - 35V	1	1000	1000	mA
	+5V Regulator	LM7805	(range)	1	1000	1000	mA
	+3.3V low-dropout	KA78RM33RTF	+5V - 20V	1	500	500	mA
Total Remaining Current Available on External Power Source 1						2500	mA

<i>External Power Source 2</i>	Component Name	Part Number	Supply	Output	Absolute	Total	Unit
Power Source 2 Selection	Battery	(full part number)	+9V	-9V	500	500	mA
Power Rails Connected to External Power Source 2	-5V Regulator	(full part number)	(range)	1	500	500	mA
Total Remaining Current Available on External Power Source 2						0	mA

Calculated Battery Life							
	Component Name	Part Number	Supply		Capacity	Required	
	Battery	(full part number)	+12V		500	500	
Battery Life						1	hours

Notes:
 External Supply Voltage should be determined by the dropout voltage for highest-voltage regulator (e.g., +14V for a +12V regulator).
 If you have multiple units in your design (e.g., a base unit and remote unit) then you need a separate power budget for each unit